SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

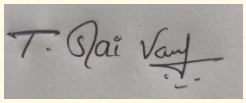
ASSESSMENT TOOL 2 – Problems

Course Code:	ITA0310	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Problems
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	Slot B	**Date:**	30-10-2026

Code of Conduct

I T. SAI VARSHINI / 192472403 (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Instructions

- This test contains 20 questions.
- Duration: 30 minutes (At home)

Annexure A - QUESTIONS

1. A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?
2. A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?
3. A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?
4. A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?
5. An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?
6. A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?
7. Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?
8. A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?
9. A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?
10. A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?
11. A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?
12. A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?
13. A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?
14. A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?
15. A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?
16. A single base station covers 2.5 km² of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?
17. An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?
18. A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?
19. A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?
20. A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?
21. A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?
22. A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?
23. A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?
24. A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?
25. A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = πR^2) and needs to blanket a 350 km² service region. Roughly how many towers will the rollout require?

1.Question: A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

Formula / Method

Paging rounds = Total number of cells ÷ Cells covered per round

Calculation

= 200 ÷ 4

= 50 rounds

Final Answer: 50 paging rounds

2.Question: A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?

Formula / Method

First calculate the offered simultaneous traffic (using the average call rate × average call duration).

Calculation

Offered traffic = 40,000 calls/min × 3 min = 120,000 Erlangs (approximately 120,000 simultaneous-call equivalents).

Excess demand = 120,000 − 100,000 = 20,000.

Blocked fraction = Excess demand ÷ Offered traffic

= 20,000 ÷ 120,000 = 0.1667

= 0.1667 × 100 = 16.67%

Final Answer: Approximately 16.67% of the offered traffic is blocked

Note: This is a capacity-overflow estimate. An exact teletraffic blocking probability (for example, Erlang B) cannot be calculated from the information given; it would require a traffic model and assumptions about call arrivals.

3.Question: A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

Formula / Method

Roaming subscribers = Total subscribers × Roaming percentage

Calculation

= 600,000 × 15/100

= 600,000 × 0.15

= 90,000 subscribers

Final Answer: 90,000 subscribers

4.Question: A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?

Formula / Method

Utilization (%) = (Current load ÷ Total capacity) × 100

Calculation

= (9,000 ÷ 12,000) × 100

= 0.75 × 100

= 75%

Final Answer: 75% utilization

5.Question: An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?

Formula / Method

Step 1: Find the total number of channels. Step 2: Divide by the reuse factor.

Calculation

48 MHz = 48,000 kHz

Total channels = 48,000 kHz ÷ 200 kHz = 240

Channels per cell = 240 ÷ 4 = 60

Final Answer: 60 channels per cell

6.Question: A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

Formula / Method

Number of channels = Total allocated bandwidth ÷ Channel spacing

Calculation

30 MHz = 30,000 kHz

= 30,000 kHz ÷ 200 kHz

= 150 channels

Final Answer: 150 channels

7.Question: Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

Formula / Method

Successful handoffs = Total attempts × Success rate

Calculation

= 2500 × 99/100

= 2500 × 0.99

= 2475 successful handoffs

Final Answer: 2,475 successful handoffs

8.Question: A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?

Formula / Method

Convert transmit power to dBm, then subtract path loss.

Calculation

Transmit power (dBm) = 10 $\log_{10}$(P in mW)

12 W = 12,000 mW

= 10 $\log_{10}$(12,000) = 40.79 dBm

Received power = Transmit power − Path loss

= 40.79 − 45 = −4.21 dBm

In milliwatts: 10^(−4.21/10) ≈ 0.38 mW

Final Answer: Approximately −4.21 dBm (about 0.38 mW)

9.Question: A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

Formula / Method

Cluster capacity = Number of cells × Users per cell

Calculation

= 7 × 900

= 6,300 users

Final Answer: 6,300 simultaneous users

10.Question: A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

Formula / Method

Slot duration = Frame duration ÷ Number of slots

Calculation

= 4.8 ms ÷ 8

= 0.6 ms

Final Answer: 0.6 milliseconds per slot

11.Question: A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?

Formula / Method

For maximum Doppler shift (motion directly toward or away from the transmitter): $f_D = (v/c) \times f$.

Calculation

Convert speed: 150 km/h × (1000/3600) = 41.67 m/s

Carrier frequency f = 1.8 GHz = 1.8×10^9 Hz

Speed of light c ≈ 3×10^8 m/s

$f_D = (41.67 \div (3 \times 10^8)) \times (1.8 \times 10^9)$

= 250 Hz

Final Answer: Maximum Doppler shift ≈ 250 Hz

Note: The actual Doppler shift depends on the direction of motion. 250 Hz is the maximum magnitude when moving directly toward or away from the transmitter.

12.Question: A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?

Formula / Method

Calculate offered simultaneous-call demand, then estimate the excess above capacity.

Calculation

Offered traffic = 70,000 calls/min × 4 min = 280,000 Erlangs (approximately 280,000 simultaneous-call equivalents).

Excess demand = 280,000 − 250,000 = 30,000.

Blocked share = 30,000 ÷ 280,000

= 0.10714 × 100

= 10.71%

Final Answer: Approximately 10.71% of the offered load is blocked

Note: This is a capacity-overflow estimate, not an exact Erlang-B blocking probability. An exact blocking probability needs additional traffic assumptions.

13.Question: A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

Formula / Method

Cluster capacity = Number of cells × Users per cell

Calculation

= 9 × 600

= 5,400 users

Final Answer: 5,400 users

14.Question: A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?

Formula / Method

Dropped calls per second = Total calls per second × Drop rate

Calculation

= 60,000 × 0.4/100

= 60,000 × 0.004

= 240 calls/second

Final Answer: 240 calls are dropped per second

15.Question: A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures −87 dBm. What path loss has the signal undergone?

Formula / Method

Path loss (dB) = Transmit power (dBm) − Received power (dBm)

Calculation

= 43 − (−87)

= 43 + 87

= 130 dB

Final Answer: 130 dB path loss

16.Question: A single base station covers 2.5 km² of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?

Formula / Method

Total coverage area = Area covered by one station × Number of stations

Calculation

= 2.5 km² × 150

= 375 km²

Final Answer: 375 km²

17.Question: An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?

Formula / Method

Step 1: Calculate total channels. Step 2: Divide by the reuse factor.

Calculation

48 MHz = 48,000 kHz

Total channels = 48,000 ÷ 200 = 240

Channels per cell = 240 ÷ 6 = 40

Final Answer: 40 channels per cell

18.Question: A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

Formula / Method

Paging cycles = Total subscribers ÷ Users processed per cycle

Calculation

= 150,000 ÷ 6,000

= 25 cycles

Final Answer: 25 paging cycles

19.Question: A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?

Formula / Method

Handoffs occur after each 3-minute interval while the call is still active.

Calculation

At 3 minutes: 1st handoff

At 6 minutes: 2nd handoff

At 9 minutes: the call ends, so no further handoff is counted.

Final Answer: 2 handoffs during the 9-minute call

Note: If the question intended a simple duration ÷ interval calculation, 9 ÷ 3 = 3 intervals. However, counting completed handoffs before the call ends gives 2 (at minutes 3 and 6).

20.Question: A receiver picks up a wanted signal at −85 dBm against a noise floor of −115 dBm. What signal-to-noise ratio, in dB, does this represent?

Formula / Method

SNR (dB) = Signal power (dBm) − Noise power (dBm)

Calculation

= −85 − (−115)

= −85 + 115

= 30 dB

Final Answer: 30 dB SNR

21.Question: A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?

Formula / Method

First calculate the aggregate user data rate.

Calculation

Total required data rate = 250 × 150 kbps

= 37,500 kbps = 37.5 Mbps

If spectral efficiency is assumed to be 1 bit/s/Hz, required bandwidth = 37.5 Mbps ÷ 1 bit/s/Hz = 37.5 MHz.

Final Answer: 37.5 MHz, assuming 1 bit/s/Hz spectral efficiency

Note: The exact radio bandwidth cannot be determined without spectral efficiency, coding, and overhead details. 37.5 Mbps is the required aggregate throughput; 37.5 MHz follows only under the stated 1 bit/s/Hz assumption.

22.Question: A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

Formula / Method

Traffic (Erlangs) = Number of subscribers × Calls per subscriber per hour × Average call duration (hours)

Calculation

Convert call duration: 2.5 minutes ÷ 60 = 1/24 hour

Traffic = 1,500 × 4 × (2.5/60)

= 1,500 × 4 × 0.041667

= 250 Erlangs

Final Answer: 250 Erlangs

23.Question: A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

Formula / Method

Total control-channel bandwidth = Number of channels × Bandwidth per channel

Calculation

= 250 × 12 kHz

= 3,000 kHz

= 3 MHz

Final Answer: 3 MHz

24.Question: A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

Formula / Method

Total throughput = Number of simultaneous connections × Average throughput per connection

Calculation

= 1,500 × 600 kbps

= 900,000 kbps

= 900 Mbps

Final Answer: 900 Mbps

25.Question: A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = πR^2) and needs to blanket a 350 km² service region. Roughly how many towers will the rollout require?

Formula / Method

Step 1: Calculate the area covered by one tower. Step 2: Divide the service region by that area and round up.

Calculation

Area per tower = $\pi R^2 = \pi \times 4^2$

$= 16\pi \approx 50.27$ km²

Number of towers = $350 \div 50.27 \approx 6.96$

Round up to the next whole tower = 7

Final Answer: Approximately 7 towers

Note: This is an ideal area-only estimate assuming perfect circular coverage without overlap, terrain effects, or coverage gaps. A practical rollout may need more towers.